

GREmLN candidate multiomics evidence

| Rank | Candidate regulator | TCR activation mRNA  | Incremental BTLA mRNA | Protein abundance    | Phosphosite abundance    | Independent association |
|------|---------------------|----------------------|-----------------------|----------------------|--------------------------|-------------------------|
| 1    | KIF22               | No qualifying change | No qualifying change  | No qualifying change | —                        | No qualifying change    |
| 2    | EZH2                | No qualifying change | No qualifying change  | No qualifying change | —                        | No qualifying change    |
| 3    | HMGN3               | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 4    | HIRIP3              | No qualifying change | No qualifying change  | No qualifying change | 2min, 24h (1 site)       | Phosphosite             |
| 5    | STAT4               | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 6    | SNAPC4              | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 7    | RBPJ                | No qualifying change | No qualifying change  | No qualifying change | —                        | No qualifying change    |
| 8    | CREM ●              | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 9    | STAU2               | No qualifying change | No qualifying change  | No qualifying change | 20min, 4h, 24h (2 sites) | Phosphosite             |
| 10   | MLX                 | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 11   | STAT1 ●             | No qualifying change | No qualifying change  | No qualifying change | —                        | No qualifying change    |
| 12   | AHR                 | —                    | —                     | —                    | —                        | —                       |
| 13   | NFIL3               | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 14   | SRP9                | No qualifying change | No qualifying change  | No qualifying change | —                        | No qualifying change    |
| 15   | MSC                 | —                    | —                     | —                    | —                        | —                       |
| 16   | GTF3C2              | No qualifying change | No qualifying change  | No qualifying change | 4h (1 site)              | Phosphosite             |
| 17   | ZBTB32 ●            | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 18   | TFDP1               | No qualifying change | No qualifying change  | No qualifying change | 4h (1 site)              | Phosphosite             |
| 19   | BCL3                | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 20   | ZNF706              | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 21   | ZNF121              | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 22   | JUNB ●              | ↑ 1h, 24h            | No qualifying change  | —                    | —                        | —                       |
| 23   | PRDM1 ●             | ↑ 4h                 | No qualifying change  | —                    | —                        | —                       |
| 24   | IRF8 ●              | No qualifying change | No qualifying change  | —                    | —                        | —                       |
| 25   | PLAGL2 ●            | No qualifying change | No qualifying change  | No qualifying change | —                        | No qualifying change    |

Contextual mRNA (qualifying)

Independent protein / phosphosite

No qualifying change

— absent / not assessable

● shared with the other model's top 25

Failed QC = quality-control failure

Candidate mRNA is contextual (TCR vs unstimulated control; BTLA+TCR vs TCR) and is shown in blue. Protein and phosphosite are independent observed molecular layers (BTLA+TCR vs TCR; source thresholds) and are shown in green. The combined BTLA+TCR vs unstimulated-control mRNA contrast was not available from the fitted source models and was not reconstructed. Direction is not interpreted as supportive or opposing (unsigned rankings). Row order matches Table 4A exactly

Figure | GREmLN top 25 candidate regulator multiomics evidence summary.